

Sushant S. Mahajan

SUMMARY

Research scientist and computational physicist with experience turning large scientific datasets into forecasting, inference, and software systems. Strong background in Python, time-series analysis, rare-event prediction, scientific ML-adjacent workflows, and cross-functional research communication.

EXPERIENCE

Postdoctoral Fellow, Stanford University 2022–present

- Lead research on large-scale solar flows using observational datasets and quantitative modeling.
- Publish first-author work on signal extraction, flow measurement, and solar-cycle-scale trends.
- Contribute to data-sharing and scientific software efforts recognized by the 2026 Stanford University Libraries Data Sharing Prize.

Solar Physics Postdoctoral Fellow, University of Hawaii 2019–2022

- Built analysis workflows for active-region inflows, meridional flow, and solar-cycle dynamics.
- Presented results at AGU, SPD/AAS, SHINE, and related research meetings.

Graduate Researcher and Teaching Assistant, Georgia State University 2014–2019

- Worked on solar flare forecasting, time-series analysis, and large-scale solar magnetogram processing.
- Mentored students and supported undergraduate physics and astronomy instruction.

SELECTED IMPACT

- First-author publications in *Solar Physics* and *The Astrophysical Journal*.
- Editor’s Choice recognition from *Solar Physics* in 2024.
- Experience with rare-event forecasting, large time-series datasets, and reproducible research software.
- Public-facing science communication through media appearances and outreach events.

SKILLS

Python, MATLAB, Fortran, bash, scientific computing, time-series modeling, data analysis, machine learning workflows, OpenMP, MPI, CUDA, visualization, technical writing, scientific presentations.

EDUCATION

Ph.D. in Astronomy, Georgia State University 2019

M.S. in Physics, Georgia State University

M.Tech. in Engineering Physics, IIT (BHU)